Performance of grid-connected PV

PVGIS-5 estimates of solar electricity generation:

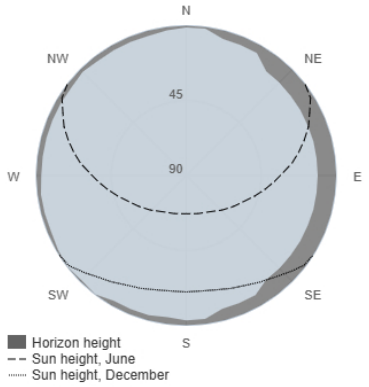
Provided inputs:

Latitude/Longitude:	46.244,6.197
Horizon:	Calculated
Database used:	PVGIS-SARAH3
PV technology:	CdTe
PV installed:	14.24 kWp
System loss:	25 %

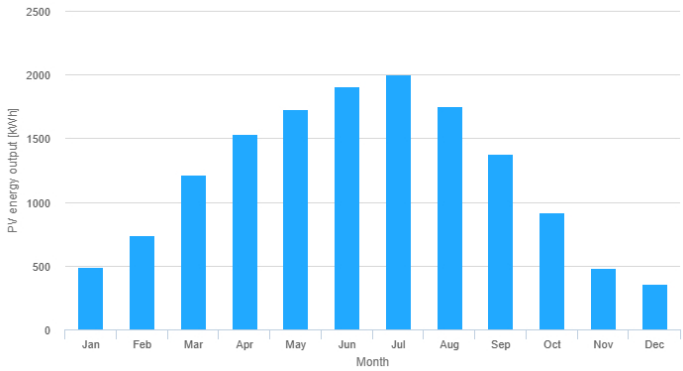
Simulation outputs

Slope angle:	10 °
Azimuth angle:	-8 °
Yearly PV energy production:	14513.36 kWh
Yearly in-plane irradiation:	1484.2 kWh/m²
Year-to-year variability:	576.14 kWh
Changes in output due to:	
Angle of incidence:	-3.31 %
Spectral effects:	1.41 %
Temperature and low irradiance:	-6.62 %
Total loss:	-31.33 %

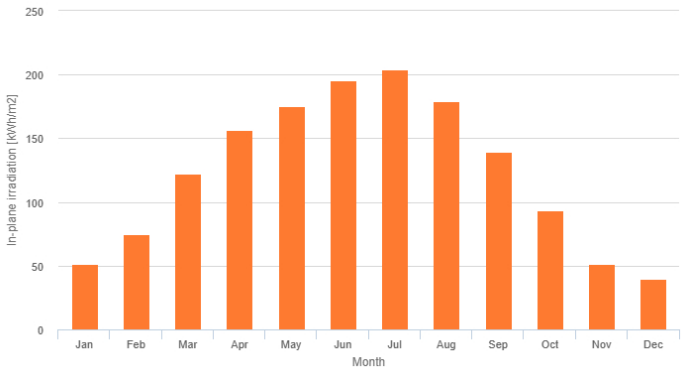
Outline of horizon at chosen location:



Monthly energy output from fix-angle PV system:



Monthly in-plane irradiation for fixed-angle:



Monthly PV energy and solar irradiation

Month	E_m	H(i)_m	SD_m
January	487.4	51.4	85.3
February	736.3	75.0	141.1
March	1216.3	122.5	183.0
April	1535.0	156.5	217.6
May	1731.9	175.2	207.8
June	1904.9	195.9	187.4
July	1998.7	204.3	209.1
August	1754.5	179.3	118.5
September	1382.0	139.4	102.7
October	922.2	93.2	121.8
November	486.3	51.7	82.7
December	357.9	39.9	56.5

E_m: Average monthly electricity production from the defined system [kWh].

H(i)_m: Average monthly sum of global irradiation per square meter received by the modules of the given system [kWh/m²].

SD_m: Standard deviation of the monthly electricity production due to year-to-year variation [kWh].

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